Full Stack Development with MERN

Project Documentation format

1. Introduction

- **Project Title:** SB Stocks
- **Team Members:**

Team Members

Name	Role
Samiksha Sahare	Frontend Developer (React UI)
Shabarish Sujith	Backend Developer (Node.js, Express API)
Ronak Jaiswar	Documentation
Sakshi Badhe	Documentation

2. Project Overview

Purpose: The **SB Stocks** platform is a virtual stock trading system designed to help users learn and practice stock trading without financial risk. It simulates real-world stock market operations where users can create portfolios, buy or sell stocks, and monitor their investment performance.

The project aims to provide a **hands-on learning environment** for understanding trading strategies, portfolio management, and market analysis.

Features:
- User Registration and Login
- Create and Manage Virtual Portfolio
- Buy and Sell Stocks
- Track Portfolio Performance
- Real-time Stock Price Simulation
- Transaction History
- Secure Authentication using JWT

3. Architecture

- **Frontend:** The frontend is built using **React.js**, providing a responsive and interactive user interface.

Technologies used:

- React.js

- React Router
- Redux (State Management)
- Axios (API Calls)
- Material UI / Bootstrap

Main responsibilities:

- User Interface Rendering
- Handling user interactions
- Sending API requests to backend
- Displaying stock data and portfolio information

• **Backend:** The backend server is developed using **Node.js and Express.js**.

Responsibilities:

- Handle API requests
- Business logic implementation
- User authentication
- Portfolio management
- Transaction processing

Technologies used:

- Node.js
- Express.js
- JWT Authentication
- bcrypt (Password encryption)

• **Database:** The application uses **MongoDB** as the database.

Main collections:

- Users
- Stocks
- Portfolios
- Transactions
- MarketData

MongoDB is used to store:

- User account details
- Portfolio information
- Stock data
- Transaction history

4. Setup Instructions

• **Prerequisites:** Install the following software:

- Node.js
- MongoDB
- Git
- VS Code

Installation:

Step 1: Clone Repository

git clone https://github.com/samikshasahare31/stock-trading.git

Step 2: Navigate to Project

cd sb-stocks

Step 3: Install Dependencies

Client:

cd client

npm install

Server:

cd server

npm install

Step 4: Setup Environment Variables

Create a `.env` file in the server folder.

5. Folder Structure

• **Client:**

```
client
│
├── src
│   ├── components
│   ├── pages
│   ├── redux
│   ├── services
│   ├── styles
│   ├── App.js
│   └── index.js
│
├── public
└── package.json
```

• **Server:**

```
server
│
├── controllers
├── models
├── routes
├── middleware
├── config
├── server.js
└── package.json
```

6. Running the Application

- **Frontend:** `npm start` in the client directory.

cd server

npm start

- **Backend:** `npm start` in the server directory.

cd client
npm start

7. API Documentation

Swagger Link:
https://stock-trading-n59r.onrender.com/api-docs/

8. Authentication

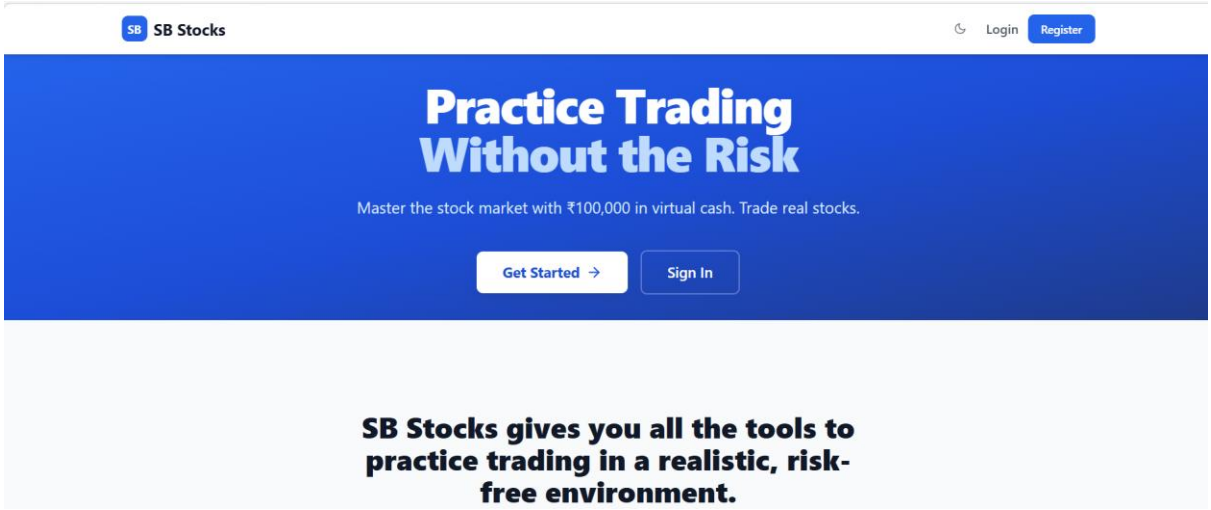
- **Authentication: JWT**-based, stateless. Users log in/register via API, receive a JWT token (expires in 7 days), stored in client localStorage. Axios interceptor adds Bearer

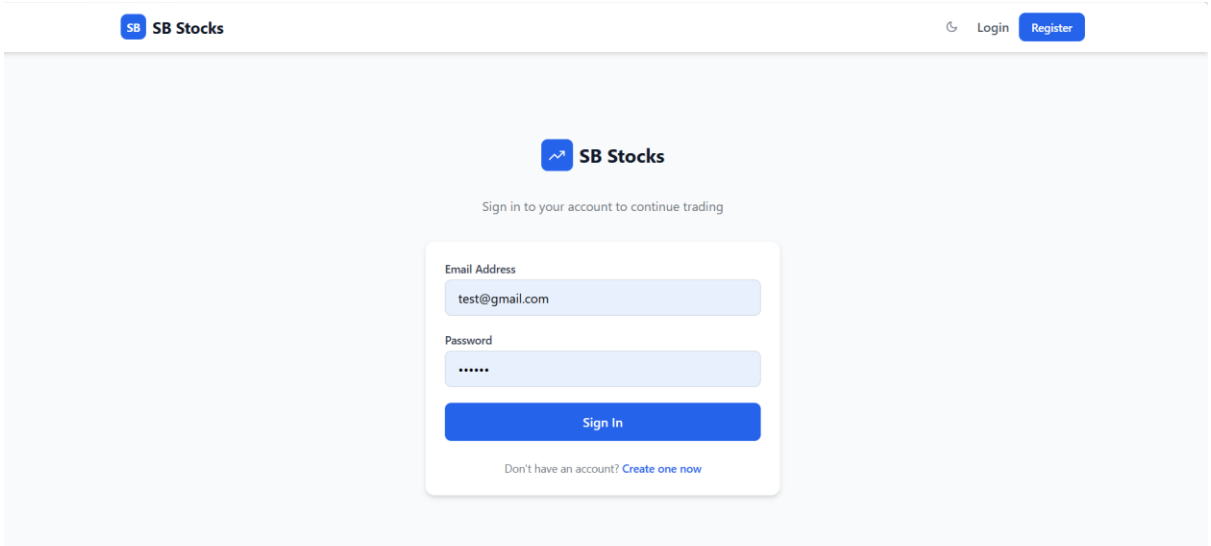
<token> to requests. Server verifies token with JWT_SECRET and attaches user to request.

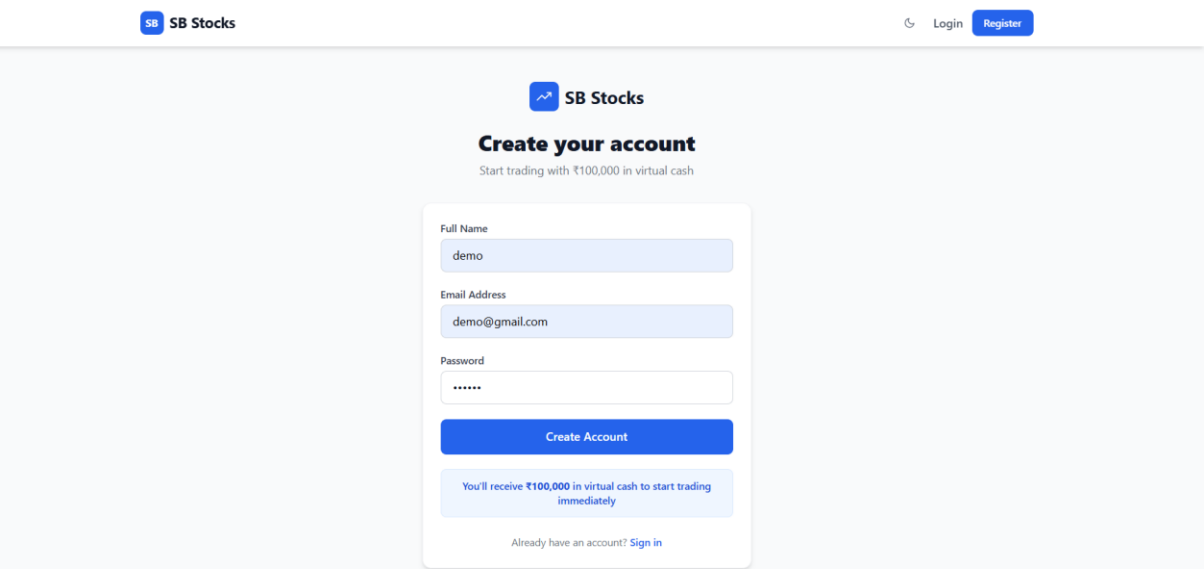
- **Authorization:** Role-based (user or admin). protect middleware checks JWT on routes; adminOnly checks role. Client-side ProtectedRoute component redirects unauthenticated users or non-admins.
- **Security:** Passwords hashed with bcrypt. No sessions; tokens handle state. Protected routes return 401/403 errors for invalid access.

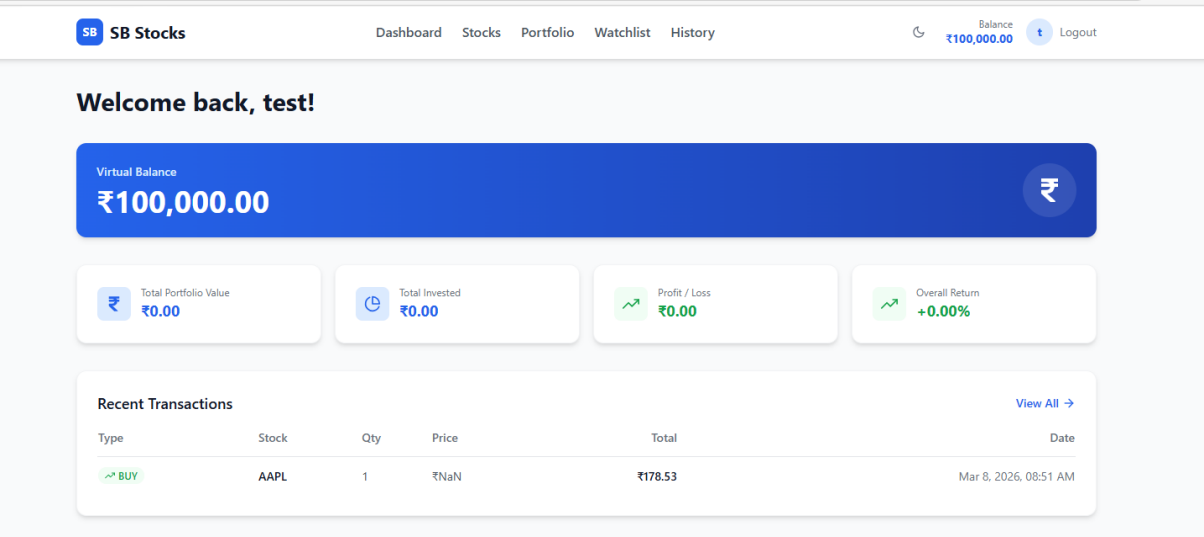
9. User Interface

Landing Page:






SB Stocks
Login
Register
SB Stocks
Create your account
Start trading with ₹100,000 in virtual cash
Full Name
demo
Email Address
demo@gmail.com
Password
••••••
Create Account
You'll receive ₹100,000 in virtual cash to start trading immediately
Already have an account? Sign in


SB Stocks
Dashboard
Stocks
Portfolio
Watchlist
History
Balance
₹100,000.00
Logout
Welcome back, test!
Virtual Balance
₹100,000.00
Total Portfolio Value
₹0.00
Total Invested
₹0.00
Profit / Loss
₹0.00
Overall Return
+0.00%
Recent Transactions
View All →
Type
Stock
Qty
Price
Total
Date
BUY
AAPL
1
₹NaN
₹178.53
Mar 8, 2026, 08:51 AM

SB Stocks

Dashboard Stocks Portfolio Watchlist History

Balance ₹100,000.00 t Logout

Portfolio

Portfolio Value	Cash Balance	Invested	Total P&L	Return
₹0.00	₹0.00	₹0.00	+₹0.00	+0.00%

Allocation

AAPL

Portfolio Breakdown

HOLDINGS	CURRENT VALUE	CASH
1	₹0.00	₹0.00

INVESTED	UNREALIZED P&L	RETURN
₹0.00	+₹0.00	+0.00%

Holdings

STOCK	QTY	AVG BUY PRICE	CURRENT PRICE	CURRENT VALUE	P&L	P&L %	ACTION
AAPL Apple Inc.	1	₹178.53	₹178.53	₹178.53	+₹0.00	+0.00%	Trade

SB Stocks

Dashboard Stocks Portfolio Watchlist History

Balance ₹100,000.00 t Log

Stock Market

Search by symbol or company name...

All Sectors

Showing 20 of 26 stocks

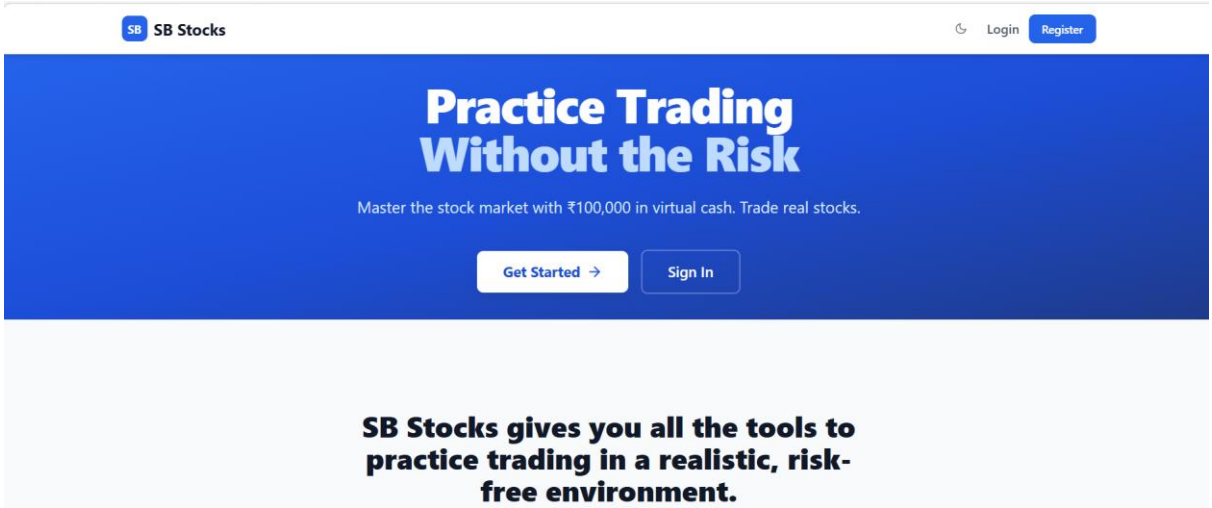
Symbol	Company Name	Price	Change	Sector	Action
AAPL	Apple Inc.	₹178.53	+0.21 (+0.12%)	Technology	View
ADBE	Adobe Inc.	₹560.16	+1.96 (+0.35%)	Technology	View
AMZN	Amazon.com Inc.	₹179.55	+1.30 (+0.73%)	Consumer	View
BAC	Bank of America Corp.	₹35.82	+0.00 (+0.00%)	Finance	View
CRM	Salesforce Inc.	₹272.45	+0.00 (+0.00%)	Technology	View
CSCO	Cisco Systems Inc.	₹52.30	+0.00 (+0.00%)	Technology	View
DIS	The Walt Disney Co.	₹111.32	-0.98 (-0.87%)	Communication	View
GOOGL	Alphabet Inc.	₹141.80	+0.00 (+0.00%)	Technology	View
HD	The Home Depot Inc.	₹345.60	+0.00 (+0.00%)	Consumer	View
INTC	Intel Corporation	₹42.18	+0.00 (+0.00%)	Technology	View

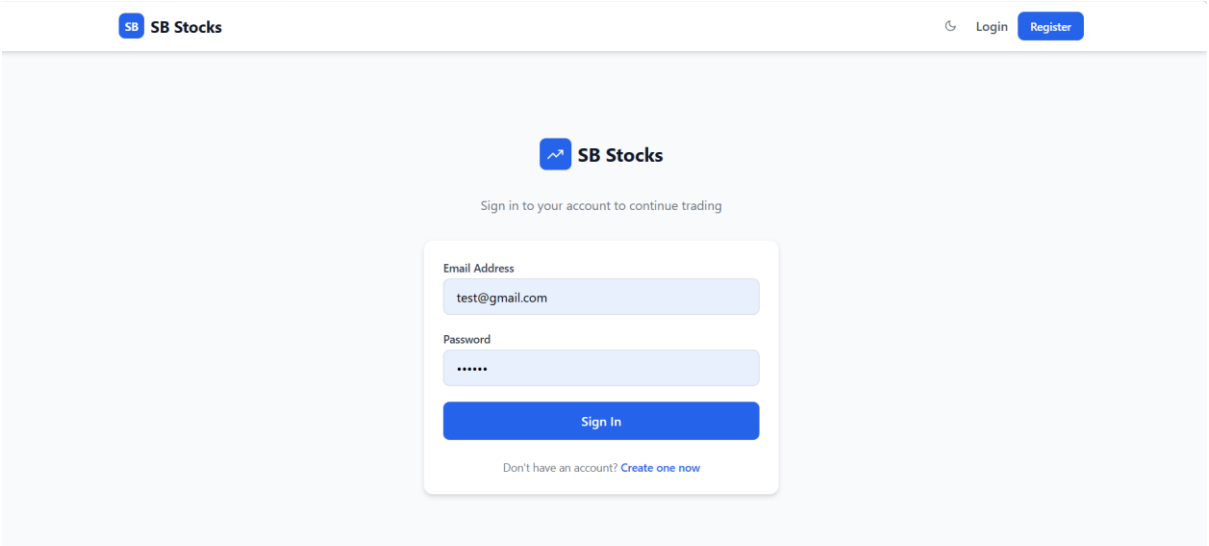
10. Testing

Testing methods used:

- Manual testing
- API testing using **Postman**

11. Screenshots or Demo





SB Stocks

Login Register

SB Stocks

Create your account

Start trading with ₹100,000 in virtual cash

Full Name

demo

Email Address

demo@gmail.com

Password

••••••

Create Account

You'll receive ₹100,000 in virtual cash to start trading immediately

Already have an account? Sign in

SB SB Stocks

Dashboard Stocks Portfolio Watchlist History

Balance ₹100,000.00 t Logout

Portfolio

Portfolio Value	Cash Balance	Invested	Total P&L	Return
₹0.00	₹0.00	₹0.00	+₹0.00	+0.00%

Allocation

AAPL

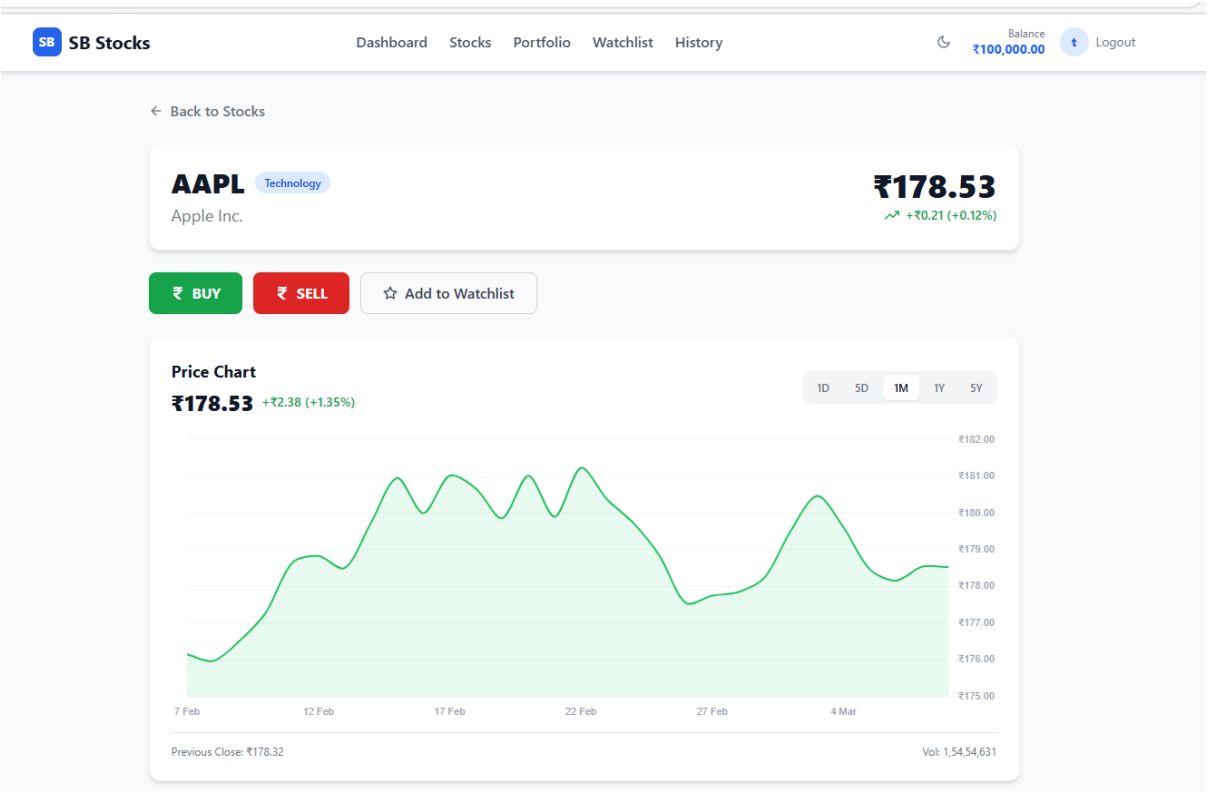
Portfolio Breakdown

HOLDINGS	CURRENT VALUE	CASH
1	₹0.00	₹0.00

INVESTED	UNREALIZED P&L	RETURN
₹0.00	+₹0.00	+0.00%

Holdings

STOCK	QTY	AVG BUY PRICE	CURRENT PRICE	CURRENT VALUE	P&L	P&L %	ACTION
AAPL Apple Inc.	1	₹178.53	₹178.53	₹178.53	+₹0.00	+0.00%	Trade


SB Stocks
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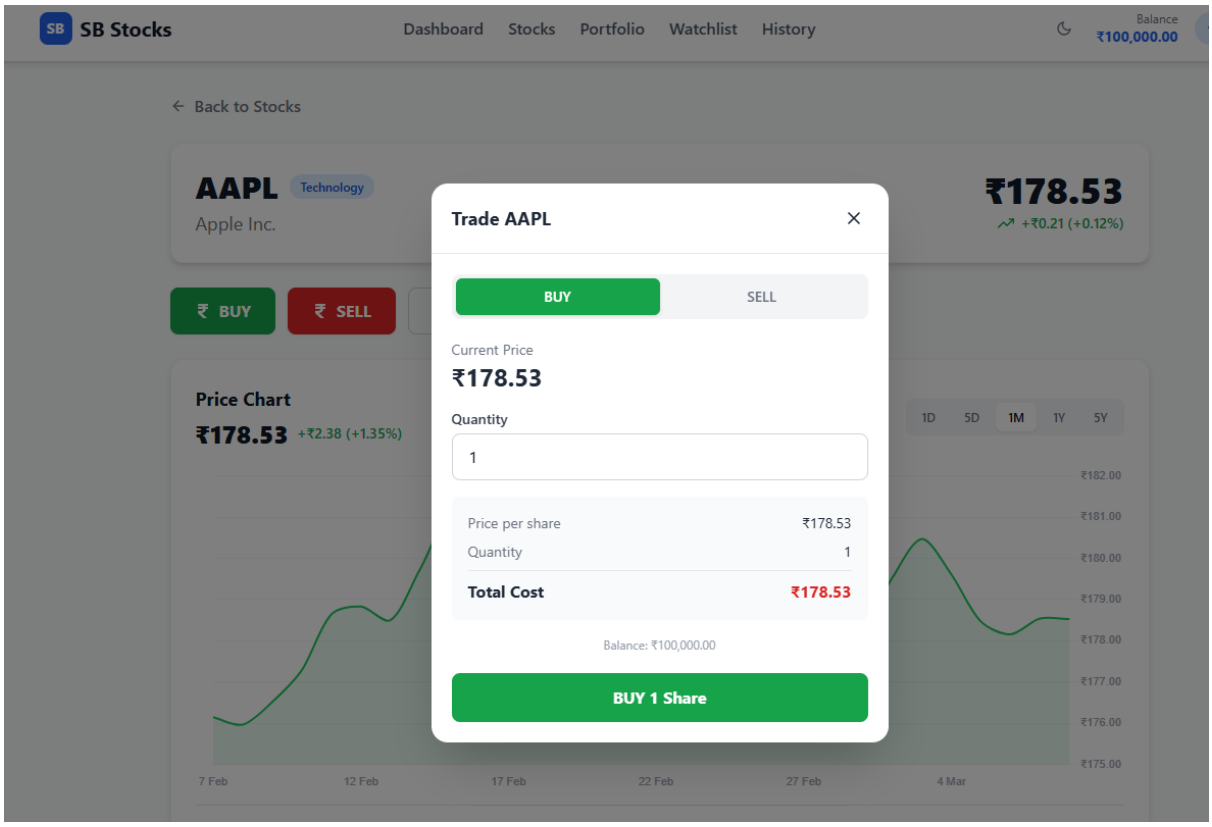
₹100,000.00
Logout
Back to Stocks
AAPL
Technology
Apple Inc.
₹178.53
+₹0.21 (+0.12%)
BUY
SELL
Add to Watchlist
Price Chart
₹178.53
+₹2.38 (+1.35%)
1D
5D
1M
1Y
5Y
₹182.00
₹181.00
₹180.00
₹179.00
₹178.00
₹177.00
₹176.00
₹175.00
7 Feb
12 Feb
17 Feb
22 Feb
27 Feb
4 Mar
Previous Close: ₹178.32
Vol: 1,54,54,631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Portfolio
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Balance
₹100,000.00
Back to Stocks
AAPL
Technology
Apple Inc.
₹178.53
+₹0.21 (+0.12%)
BUY
SELL
Price Chart
₹178.53
+₹2.38 (+1.35%)
1D
5D
1M
1Y
5Y
Trade AAPL
BUY
SELL
Current Price
₹178.53
Quantity
1
Price per share
₹178.53
Quantity
1
Total Cost
₹178.53
Balance: ₹100,000.00
BUY 1 Share
7 Feb
12 Feb
17 Feb
22 Feb
27 Feb
4 Mar

12. Known Issues

- JWT Security Limitations: Tokens are stored in localStorage, which is vulnerable to XSS attacks. No refresh token mechanism exists, so users must re-login after token expiration (7 days). Consider using httpOnly cookies for production.
- Error Handling Gaps: While basic error responses are implemented, some edge cases (e.g., network failures, database timeouts) may not be handled gracefully, potentially leading to unhandled promises or unclear user feedback.

13. Future Enhancements

- Enhanced Security: Add refresh tokens, implement OAuth (e.g., Google login), use httpOnly cookies for tokens, and add CSRF protection. Integrate security scanning tools.
- Real-Time Features: Use WebSockets (e.g., Socket.io) for live stock price updates. Integrate with real market APIs (e.g., Alpha Vantage, Yahoo Finance) for accurate data.